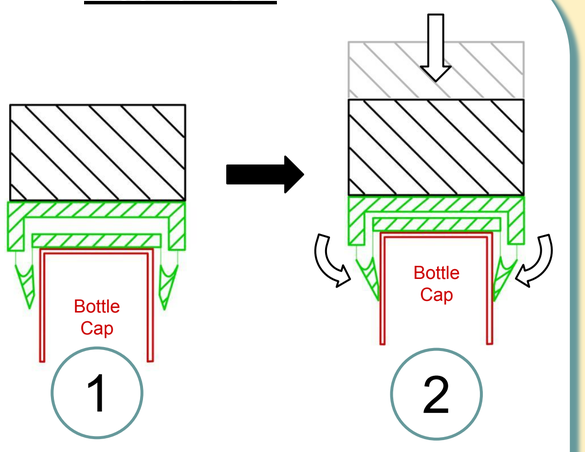
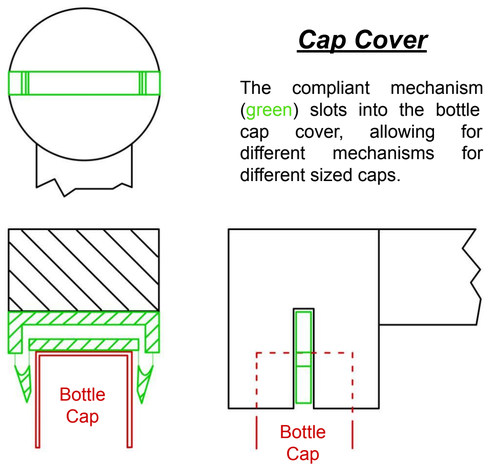
DESIGN IDEATION AND CONCEPT GENERATION

Mechanism



Bottle is placed inside the bottle cap cover

Cap pushes into cap cover and clamps move in to grab the bottle cap



Cap Cover

The compliant mechanism (green) slots into the bottle cap cover, allowing for different mechanisms for different sized caps.

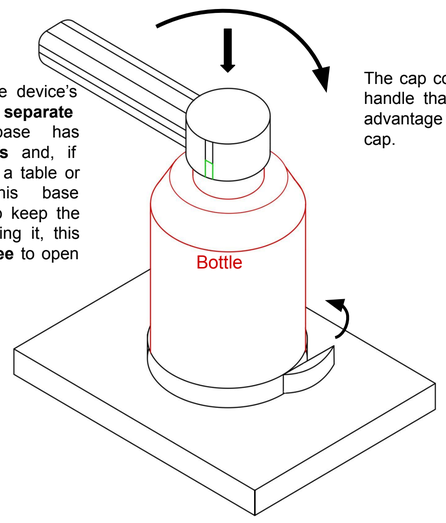


- The claw clamps the bottom of the cap and the plate acts on the top providing force when the handles are pressed.
- The handles have cut-outs to grip and is covered by an ergonomic foam.
- For mechanical advantage, the handle is long to increase moment arm when force is applied.

- Belt like mechanism to hold the bottle/jar at the base.
- Buckle to tighten at different diameters.

Full Assembly

The Bottle **clamps** the device's base, which is a **separate component**. The base has **anti-slip rubber pads** and, if needed, can clamp to a table or kitchen counter. This base eliminates the need to keep the bottle firm while opening it, this **leaves both hands free** to open the bottle.



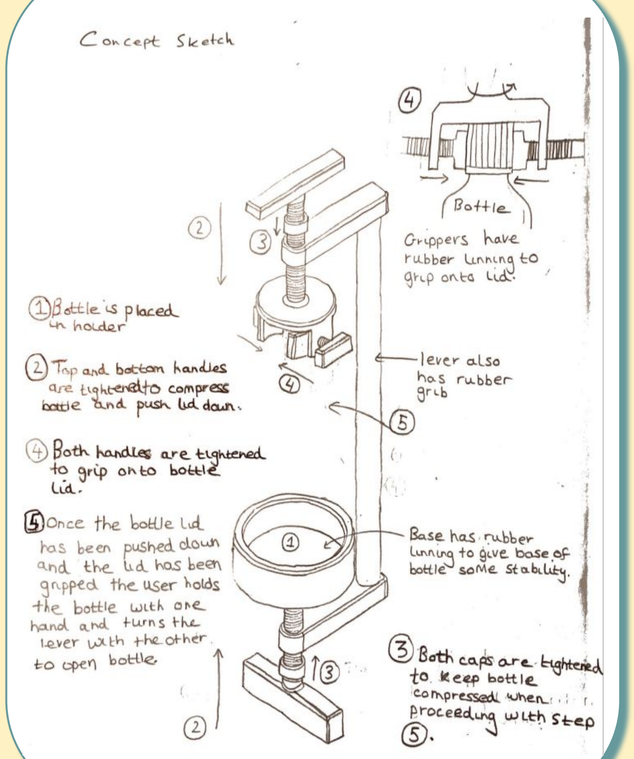
The cap cover has an ergonomic handle that provides mechanical advantage when turning the bottle cap.

Concept 1: Compliant Mechanism and retaining base

- Easy to use/ manufacture.
- Can be used with caps and bases of different diameters.
- The top handle can be difficult to turn as a force still needs to be applied.
- Execution of the idea may be more difficult

Concept 2: Belt and holder mechanism

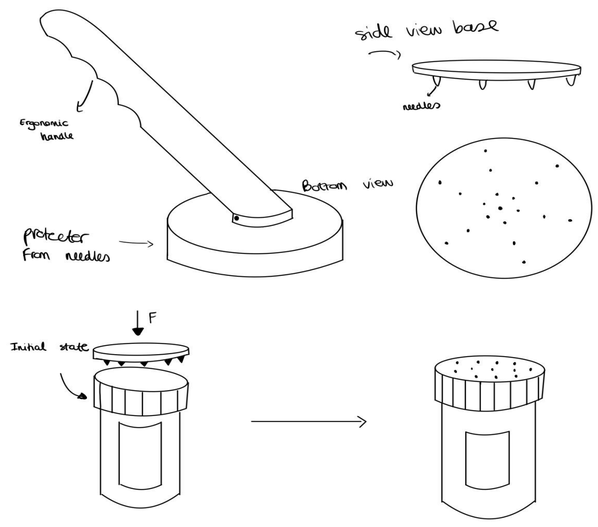
- Easy to manufacture.
- Simple and intuitive to use.
- Belt can tighten or loosened to hold bottles of different diameters.
- Lightweight and has ergonomic foam to hold the handles.
- Takes up little space.
- The "claw" might not necessarily hold on to thin caps.
- The top handle will be difficult to grip.
- Requires both hands to open.



Concept #3: Clamp holder and bottle opener

- Easy opening when the mechanism is set up
- Reliable
- Secure grip

- Multi-step process not simple to use
- High forces needed to set-up, compression and clamping required
- Heavy
- Difficult to manufacture
- Difficult to assemble
- Not intuitive to use
- Limited to certain height bottles



- By making holes at the cap, the pressure difference between the canister and the outside environment is reduced, therefore reducing the pressure on the cap and the force required to open it.

Concept 4: Pressure difference removal by hole punching

- Easy to manufacture
- Works for a wide range of caps
- Takes up little space
- Light
- Only one component
- Can be used for different cap sizes

- Breaks the mechanism, the cap is no longer child proof
- Contains sharp edges
- Hard force required

MCDA decision matrix

Criteria		Concept 1 - Luca			Concept 2 - Nida			Concept 3 - Nathan			Concept 4 - Lucia		
		Weight	*Score*	*Weighted Score*	*Weight*	*Score*	*Weighted Score*	*Weight*	*Score*	*Weighted Score*	*Weight*	*Score*	*Weighted Score*
	1) Performance												
1)	The bottle opener shall be simple and intuitive to use requiring little force to operate	4	3	12	4	3	12	4	2	8	4	3	12
2)	The system shall work for bottles with different sizes	5	4	20	5	4	20	5	5	25	5	5	25
3)	The bottle opener shall be available at an accessible price point	2	4	8	2	5	10	2	3	6	2	4	8
	2) Geometry												
4)	The bottle opener shall have a mass no more than 700 grams	3	3	9	3	5	15	3	2	6	3	5	15
5)	The system shall minimize the number of components	1	3	3	1	4	4	1	1	1	1	5	5
6)	The bottle opener shall be easy to store	3	5	15	3	4	12	3	3	9	3	4	12
	3) Safety												
7)	The mechanism should not have sharp edges	2	5	10	2	4	8	2	5	10	2	3	6
8)	If the product fails/is dropped, it does not cause serious injuries	4	3	12	4	4	16	4	3	12	4	4	16
9)	The holding system should not be stiff when used, so as to cause pain/injury.	5	4	20	5	2	10	5	3	15	5	2	10
Total		109			107			92			109		